

his alma mater, Tulane University. Mr. N's financial advisor, Mr. Clayton Spencer, has been working with Mr. N for less than a year. During that time, Spencer has proposed a comprehensive financial plan. Despite Spencer's recommendations, however, Mr. N's asset allocation has remained the same, at nearly 80 percent equities, with 35 percent in his employing company's publicly traded stock. Still, Spencer has developed a good working relationship with Mr. N.

Spencer believes that Mr. N. is a well-grounded, fairly rational person, but he also believes that Mr. N has some behavioral issues to address. In Spencer's view, the most important issue is that Mr. N has not taken action yet on the new, more conservative allocation that Spencer proposed months ago of 50 percent stocks, 40 percent bonds, and 10 percent cash. Spencer worries about the lack of diversification in Mr. N's portfolio. Spencer's concern is that a severe downward market fluctuation or drop in Mr. N's employing company's stock may cause him to sell assets irrationally, affecting his long-term financial plan. Spencer's financial plan demonstrates that even with a somewhat less aggressive portfolio, Mr. N could still meet his primary financial objectives if he could save just \$25,000 annually. Spencer believes that one of the issues is that Mr. N thinks of himself as a good investor because of the biotechnology bets that worked out well for him in the past. Spencer suspects that Mr. N hasn't changed his allocation because he thinks Spencer's allocation recommendation is too conservative. Spencer also notices that Mr. N constantly worries about missing out on hot stocks that go up, which his friends are investing in. Spencer decides to ask Mr. N to take a behavioral bias diagnostic questionnaire. When Spencer gets the answers to the questionnaire, he decides to focus on three biases: *regret aversion*, *overconfidence*, and *self-control*. Spencer asks Mr. N further questions on these three biases. Table 25.2 shows Mr. N's answers to these questions in **bold**.

Through this process, Spencer finds that Mr. N is indeed susceptible to the following emotional biases:

- *Regret-aversion bias* (the tendency to avoid making a decision for fear the decision may cause regret later).
- *Overconfidence bias* (the tendency to overestimate one's investment savvy).
- *Self-control bias* (the tendency to spend today rather than save for tomorrow).

As part of the original financial planning process, Spencer administered a risk tolerance questionnaire to Mr. N for the purpose of generating a mean-variance optimization portfolio recommendation. When Spencer generated